

Dress the Bed Pillows and Throw SOP (1x Episode: two pillows and one throw, in situ)

One episode dresses the head of the bed and the foot of the bed where the bed stands. The first pillow goes into its case, is plumped, and is stood at the headboard. Then the second pillow goes into its case, is plumped, and is stood beside the first. Then the throw is folded in thirds and draped across the foot. The episode ends when both pillows stand at the headboard and the folded throw lies square across the foot. Never split the pillows and the throw into two episodes.

The base is **passive**: it has no drive of its own, so it is pushed by hand up to the long side of the bed and locked there, and nothing is carried out of the room. Everything the episode touches is already on the bed when recording starts: the made bed, the two bare pillows, the two folded cases, and the folded throw.

The bed is worked **as found**. The sheet and the blanket are already flat and tucked. Nothing pulls them, straightens them, or tucks them again. The **headboard stands at the left end** of the bed as the arms see it and the **foot is at the right end**, so the **left half** of the bed top is the head half and the **right half** is the foot half.

This is an in-situ task, and four things follow from that. First, **nothing is ever dropped onto the bed**: a thing is brought down onto the bed and let go only when it is flat and in its place. Second, **the bedding under the work never moves**: nothing drags, pulls, or bunches the sheet or the blanket. Third, **the head of the bed is worked first and the foot second**, and nothing goes back to the head once the throw is opened. Fourth, **only one thing moves at a time**, and nothing is passed from one gripper to the other.

Nothing goes on the floor. A pillow, a case, or the throw that touches the floor is a drop. It is not picked up and the episode does not carry on. It is reported.

The bed top is set up in one of three ways. The bed, the headboard, the cases, and the throw never move. Only the two bare pillows do.

- **Config L**: the two bare pillows lie in the **head half**, along the front edge, with the case mouths pointing at them, toward the head.
- **Config M**: the two bare pillows lie at the **front of the middle**, along the front edge just in front of the cases, with the case mouths pointing toward the foot.
- **Config R**: the two bare pillows lie in the **foot half**, along the front edge, with the case mouths pointing at them, toward the foot.

One config per episode, chosen before recording and never changed mid-episode. Where a step depends on the setup it says so on an **IF** line. Look at the bed and follow the line that matches.

Same-side rule: the gripper on the pillows' side is the one that feeds each pillow into its case. The other gripper holds the case mouth open. In Config M the pillows lie in the shared middle and the right gripper feeds. After that the roles are fixed: the **left gripper** carries every cased pillow to the headboard and stands it up, because the headboard is on its side, and the **right gripper** leads the work on the throw at the foot, with the left gripper taking the far end of every fold and carry.

The order never changes: **case the first pillow, plump it, stand it at the headboard; case the second pillow, plump it, stand it beside the first; fold the throw in thirds; drape it across the foot; square it.**

Setup

Complete the base positioning and all the checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own, it is pushed into place by hand, and it is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the near long side of the bed, with the **headboard to the left** and the **foot to the right**. Stop it **square to the side edge**, so the side edge runs straight across the frame and neither end of the bed sits nearer than the other.
2. Stop it **centered on the long side**, so the headboard and the foot are the same distance from the middle of the base.
3. Stop it **close enough** that either gripper reaches the far edge of the bed top at the middle without the arm extending, and **far enough** that no arm, wrist, or part of the base touches the bed frame, the headboard, or the wall while both arms work.
4. Check the **head**: with the base parked, the **left gripper** reaches the headboard end, holds a pillow upright against the headboard at the back half and at the front half, and lets go, without the wrist or the forearm touching the headboard or the wall.
5. Check the **foot**: the **right gripper** reaches the foot end and the front hanging corner of a throw draped there, and the **left gripper** reaches the far end of a folded throw lying across the foot, both without extending.

6. Check the **bed top**: the gripper on the pillows' side reaches both bare pillows where this config puts them, both grippers reach the cases at the middle, and both reach the folded throw at the back corner of the foot end, all without extending and without dragging the bedding.
7. Lock or brake the base. Push it firmly once by hand: it must not roll, creep, or turn.
8. If any of lines 1 to 6 fails, push the base to a new park by hand and start again at line 1. Do not work a bed the arms cannot reach comfortably.

The base stays locked and still for the whole episode. Nothing moves it: no arm leans on the bed or the headboard hard enough to shift it, nothing and nobody touches it, and it is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the middle of the bed top and its frame includes the whole bed top, the headboard, the foot end with both side edges, the two bare pillows, the two folded cases, and the folded throw.
3. The camera reads the bed from the side, so where a thing lies and how square it is are readable.
4. The camera reads the headboard, so a pillow standing against it and a pillow leaning away from it can be told apart.
5. The camera reads the foot end and both side edges there, so whether the throw hangs the same on both sides is readable.
6. The camera reads the top face of a pillow and the top face of the throw, so creases, lumps, and an open case mouth are readable.
7. Both arms are at home with grippers open.
8. The **left arm** reaches the head half of the bed top, the headboard end, the cases at the middle, and the far end of the throw, without extending to a joint limit.
9. The **right arm** reaches the foot half of the bed top, the cases at the middle, the folded throw at the back corner of the foot end, and the front hanging corner of the draped throw, without extending to a joint limit.
10. The gripper on the pillows' side reaches both bare pillows where this config puts them without extending.
11. Every lift comes just off the bed and every carry stays low and level over the bed top. No arm needs to lift a thing high or carry it over the headboard.
12. The two arms do not collide. The arm carrying a pillow to the headboard stays out of the other arm's path along the bed.
13. If a place cannot be reached, re-park the base by the Base positioning steps until lines 8 to 12 hold.

Materials checklist

1. The **bed** is made. The sheet and the blanket are flat, tucked, and dry. Nothing straightens, pulls, or tucks them again.
2. The **headboard** stands fixed at the left end of the bed. It does not rock. Nothing is hung on it, leaned on it, or pushed against it hard.
3. The bed top is clear apart from the two bare pillows, the two folded cases, and the folded throw.
4. There are exactly **two bare pillows**, the same size and the same shape. Each is soft enough to press and springs back when let go. Neither is torn, damp, or stained.
5. The two bare pillows lie flat at the **pillow spot** this episode's config puts them, one on top of the other, long edges running along the bed: the **head half in Config L**, the **front of the middle in Config M**, the **foot half in Config R**.
6. There are exactly **two cases**, the same size and the same colour, and both fit the pillows. Each is clean, dry, and right side out.
7. The two cases lie folded flat at the **case spot**, the middle of the bed top, one on top of the other, with their **mouths pointing at the pillow spot**: toward the head in Config L, toward the foot in Config M and R.
8. There is **one throw**. It is a flat cloth, longer than the bed is wide, so that draped across the foot it hangs the same over the front edge and the back edge. It is clean, dry, and not creased into hard folds.
9. The throw lies **folded in half once** at the **throw spot**, the back corner of the foot end, out of the way of the pillows, with its folded edge toward the head.
10. The foot of the bed is clear. Nothing stands on it or against it.
11. The floor around the bed is clear and dry, so an arm has room to work at both ends and nothing that is dropped rolls under the base.

Workspace layout

Nothing is marked or taped out. Every place below is judged by eye from the bed itself.

- **Bed top:** the flat made bed. All the work happens on it. The bedding under the work never moves.
 - **Head half:** the left half of the bed top, toward the headboard. The left gripper's side.
 - **Foot half:** the right half of the bed top, toward the foot. The right gripper's side.
- **Headboard end:** the left end of the bed, at the headboard. Both cased pillows end up standing here, side by side: the first at the **back half** of the headboard, the second at the **front half**.
- **Foot end:** the right end of the bed. The folded throw ends up lying across here, square.

- **Pillow spot:** where the two bare pillows lie at the start, along the front edge, stacked: head half in Config L, front of the middle in Config M, foot half in Config R.
- **Case spot:** the middle of the bed top, where the two folded cases lie stacked, mouths toward the pillow spot. Every pillow is cased and plumped here.
- **Throw spot:** the back corner of the foot end, where the folded throw lies at the start.
- **Headboard:** fixed. Nothing is ever hung on it, leaned on it, or pushed against it hard.

Arm assignments

- **The gripper on the pillows' side is the feed gripper** for both pillows: the left gripper in Config L, the right gripper in Config M and R. It lifts each bare pillow and pushes it into its case, then pulls the case mouth back over it.
- **The other gripper is the mouth gripper** for both pillows: the right gripper in Config L, the left gripper in Config M and R. It holds the case mouth open, then presses the far corners of the pillow into the closed corners of the case.
- **The left gripper owns the headboard.** It carries every cased pillow from the case spot to the headboard and stands it up. The right gripper never carries a pillow to the headboard.
- **The right gripper leads the throw.** It opens the throw, takes the front corner of every fold, the front end of the carry, and does the squaring. The **left gripper takes the back corner** of every fold and the back end of the carry.
- Both grippers plump together, one on each short end of the pillow.
- A thing is released only in its place: a cased pillow flat on the case spot, a standing pillow leaning on the headboard, the throw flat across the foot. Never open a gripper over open bed with a thing still hanging from it.

Vocabulary

- **In situ:** the bed is worked where it stands. Nothing is carried out of the room, and nothing about the room is rearranged for the task.
- **Passive base:** the base has no drive. It is pushed by hand to its park before recording, locked there, and it does not move at all during the episode.
- **Config:** which of the three pillow spots this episode uses. It is set before recording and never changes mid-episode.
- **Feed gripper:** the gripper on the pillows' side, which feeds each pillow into its case. Left in Config L, right in Config M and R.
- **Mouth gripper:** the other gripper, which holds the case mouth open. Right in Config L, left in Config M and R.
- **Bare pillow:** a pillow with no case on it.
- **Case:** the cloth bag the pillow goes into. It is closed on three sides and open at one short end.

- **Case mouth:** the open short end of the case. The pillow goes in here.
- **Mouth open:** the mouth gripper has pinched the top layer of the case at the middle of the mouth and lifted it just off the bottom layer, so the mouth stands open as a gap with the bottom layer still flat on the bed.
- **Cased:** the pillow is all the way inside the case, and the case is pulled back over it so no bare pillow shows at the mouth.
- **Corners filled:** the two far corners of the pillow sit right down in the two closed corners of the case, so the case has no empty flat ear at that end.
- **Plump:** the cased pillow lies flat on the case spot, and the two short ends are pressed in toward the middle and let go, so the filling spreads and the pillow comes up full and even. Plumping is only ever done on the bed.
- **Full and even:** the pillow is the same thickness across its middle. No hollow in the middle. No hard lump at one end.
- **Just off:** lifted only far enough that the thing no longer touches what it was lying on, and no higher.
- **Low carry:** the thing is carried just off the bed top, level, along the bed. It never goes high and never over the headboard.
- **Standing at the headboard:** the cased pillow stands upright with its bottom long edge resting on the bed, its top long edge running from the front of the bed to the back, and its face leaning on the headboard, and it does not fall forward when the gripper lets go.
- **Side by side:** the two pillows stand at the headboard touching along their short edges, the first at the back half and the second at the front half, both the same height, with the same gap to the back edge and to the front edge of the bed.
- **In thirds:** the throw is folded twice, so it ends up three layers thick and one third as wide as it started. The two folds run the same way and both edges land flat.
- **Draped at the foot:** the folded throw lies flat across the foot of the bed, running from the front edge to the back edge, hanging the same over both.
- **Square:** the throw's long edge runs parallel to the foot edge of the bed, and the same amount of throw hangs over the front edge and over the back edge.
- **Flat:** no fold, no rumple, and no lifted corner on the top face.
- **Drop:** a pillow, a case, or the throw touches the floor. It is not picked up. It is reported.
- **Bedding pulled:** the sheet or the blanket is dragged, bunched, or untucked by the work. This is a fault, not a thing to fix on camera.
- **Back and clear:** the arm is drawn back so that no part of it is over the bed top or the headboard.

Handling standard

- Move exactly one thing at a time. Nothing is passed from one gripper to the other.

- Every lift comes straight up, just off the bed, before the thing moves anywhere.
- Every carry is a low carry over the bed top.
- Nothing is ever dragged along the bedding.
- Nothing is put down anywhere but its own place in the step being done.
- Nothing rests on the headboard. No gripper, wrist, or forearm leans on it, and no pillow is pushed against it hard enough to rock it.

Steps

Steps 1 and 4 depend on the config: the gripper on the pillows' side feeds the pillow while the other holds the case mouth. Every other step is the same in all three.

Step 1: Case the first pillow

Goal: the first pillow is all the way inside its case at the case spot, with both far corners filled.

1.1 Open the case mouth

- **IF Config L:** the **right gripper** is the mouth gripper and the **left gripper** is the feed gripper.
- **IF Config M or R:** the **left gripper** is the mouth gripper and the **right gripper** is the feed gripper.

Then, in all three:

- The **mouth gripper** comes down onto the top case at the case spot and pinches the top layer of the case at the middle of the mouth.
- Lift the top layer just off the bottom layer and hold it there, so the mouth stands open. The bottom layer and the closed end of the case stay flat on the bed.

Check: the mouth is open wide enough to take a pillow, and the rest of the case lies flat. If the case is inside out, the mouth gripper lays the top layer down, turns the case right side out on the bed, and opens the mouth again.

1.2 Feed the pillow in

- The **feed gripper** comes down onto the pillow spot and pinches the top pillow at the middle of the short end nearer the case.
- Lift it just off the pillow under it and bring it in a low carry to the open mouth. Do not lift it high. Do not drag it across the bed.
- Push the pillow in through the mouth, straight down the case, until its far end reaches the closed end of the case. Then open the feed gripper and draw it out of the mouth.

Check: the pillow is all the way in and no part of it sticks out of the mouth. If it stops short, the feed gripper pinches the near end again and pushes it in further.

1.3 Fill the corners and settle the case

- The **mouth gripper** lets the top layer down, so the cased pillow lies flat on the case spot.
- The **mouth gripper** presses each far corner of the pillow down into its own closed corner of the case, the back corner first, then the front corner.
- The **feed gripper** pinches the top layer at the middle of the mouth and pulls the mouth back over the pillow, so the case lies smooth and no bare pillow shows. Then it opens and comes back and clear.

Check: the pillow is cased and both far corners are filled. The case is smooth on the top face. If a corner is empty, the mouth gripper presses that corner down again.

Expected state: the first pillow lies flat on the case spot, cased. The second pillow lies at the pillow spot and the second case lies under the first pillow's place on the case spot, both untouched. The throw has not been touched. Both grippers are back and clear.

Step 2: Plump the first pillow

Goal: the first cased pillow is full and even.

2.1 Press the ends

- Keep the pillow flat on the case spot.
- The **left gripper** closes and presses the head-end short end of the pillow in toward the middle while the **right gripper** presses the foot-end short end in toward the middle. Then both let go.
- Do this two times. Do not punch it. Do not lift it off the bed.

2.2 Even it out

- The **right gripper** presses the middle of the top face down once and lets it come back up.
- The **right gripper** presses any hard lump flat with the closed face of the gripper.

Check: the pillow is full and even. No hollow in the middle. No hard lump at one end. The bedding under the pillow has not been dragged. If it is not even, both grippers press the ends once more.

Expected state: the first pillow lies flat on the case spot, cased, full and even. Both grippers are back and clear.

Step 3: Stand the first pillow at the headboard

Goal: the first pillow stands upright against the back half of the headboard.

3.1 Lift and carry it to the head

- The **left gripper** pinches the pillow at the middle of its near long edge.

- Lift it just off the bed and carry it in a low carry along the bed top to the headboard end. Do not lift it high. Do not carry it over the headboard.

3.2 Stand it up

- At the headboard end, turn the arm so the **left gripper** points at the headboard. The pillow now hangs with its top long edge running from the front of the bed to the back.
- Lower it until its bottom long edge rests on the bed at the **back half** of the headboard and its face leans on the headboard, leaving room in front of it for the second pillow.
- Open the gripper and draw it back.

Check: the pillow is standing at the headboard. It stands upright, leans on the headboard, and does not fall forward when the gripper lets go. If it slides down or falls forward, the **left gripper** pinches its top long edge again, stands it up, and sets it down again.

Expected state: the first pillow stands at the back half of the headboard. The second pillow lies at the pillow spot and the second case lies flat on the case spot. The throw has not been touched.

Step 4: Case the second pillow

Goal: the second pillow is all the way inside its case at the case spot, with both far corners filled.

4.1 Open the case mouth

- **IF Config L:** the **right gripper** is the mouth gripper and the **left gripper** is the feed gripper.
- **IF Config M or R:** the **left gripper** is the mouth gripper and the **right gripper** is the feed gripper.

Then, in all three:

- The **mouth gripper** pinches the top layer of the second case at the middle of the mouth, lifts it just off the bottom layer, and holds the mouth open. The bottom layer and the closed end stay flat.

Check: the mouth is open wide enough to take a pillow, and the rest of the case lies flat. The first pillow at the headboard has not been knocked. If the case is inside out, the mouth gripper lays the top layer down, turns the case right side out on the bed, and opens the mouth again.

4.2 Feed the pillow in

- The **feed gripper** pinches the second pillow at the middle of the short end nearer the case, lifts it just off the bed, and brings it in a low carry to the open mouth.
- Push the pillow in through the mouth, straight down the case, until its far end reaches the closed end. Then open the feed gripper and draw it out of the mouth.

Check: the pillow is all the way in and no part of it sticks out of the mouth. If it stops short, the feed gripper pinches the near end again and pushes it in further.

4.3 Fill the corners and settle the case

- The **mouth gripper** lets the top layer down, then presses each far corner of the pillow down into its own closed corner of the case, the back corner first, then the front corner.
- The **feed gripper** pinches the top layer at the middle of the mouth and pulls the mouth back over the pillow, so the case lies smooth and no bare pillow shows. Then it opens and comes back and clear.

Check: the pillow is cased and both far corners are filled. The case is smooth on the top face. If a corner is empty, the mouth gripper presses that corner down again.

Expected state: the second pillow lies flat on the case spot, cased. The first pillow still stands at the headboard. The throw has not been touched.

Step 5: Plump the second pillow

Goal: the second cased pillow is full and even.

5.1 Press the ends

- Keep the pillow flat on the case spot.
- The **left gripper** presses the head-end short end in toward the middle while the **right gripper** presses the foot-end short end in. Then both let go.
- Do this two times. Do not punch it. Do not lift it off the bed.

5.2 Even it out

- The **right gripper** presses the middle of the top face down once and lets it come back up, then presses any hard lump flat with the closed face of the gripper.

Check: the pillow is full and even. No hollow in the middle. No hard lump at one end. The bedding under the pillow has not been dragged. If it is not even, both grippers press the ends once more.

Expected state: the second pillow lies flat on the case spot, cased, full and even. Both grippers are back and clear.

Step 6: Stand the second pillow at the headboard

Goal: both pillows stand side by side at the headboard.

6.1 Lift and carry it to the head

- The **left gripper** pinches the pillow at the middle of its near long edge, lifts it just off the bed, and carries it in a low carry along the bed top to the headboard end.

- Do not lift it high. Do not carry it over the headboard. Do not knock the first pillow on the way.

6.2 Stand it up beside the first

- At the headboard end, turn the arm so the **left gripper** points at the headboard, so the pillow hangs with its top long edge running from the front of the bed to the back.
- Lower it until its bottom long edge rests on the bed at the **front half** of the headboard, its face leaning on the headboard and its back short edge touching the first pillow's front short edge.
- Open the gripper and draw it back.

Check: the two pillows are side by side. Both stand upright against the headboard, both are the same height, their short edges touch, and the gap to the back edge of the bed and to the front edge is the same. Neither falls forward when the gripper lets go. If one leans or sits low, the **left gripper** pinches its top long edge again, stands it up, and sets it down again.

Expected state: both pillows stand cased and plumped at the headboard. The bed top from the pillows to the foot is clear except for the folded throw. Both grippers are back and clear.

Step 7: Fold the throw in thirds

Goal: the throw lies at the foot half three layers thick and one third as wide as it started, running from the front edge of the bed to the back edge.

7.1 Open the throw flat

- The **right gripper** pinches the front corner of the throw's top layer at its free edge, and the **left gripper** pinches the back corner of the same edge.
- Lift that edge just off the bed and carry it toward the head until the throw opens out flat over the foot half, its long edges running from the front edge of the bed to the back edge and hanging a little over both.
- Lay it down and let go. The **right gripper** smooths the top face once with the closed face of the gripper, from the head side to the foot side.

Check: the throw lies flat. No fold. No rumple. No lifted corner. The bedding under it has not been dragged. If a corner is folded under, the gripper on that side lifts the corner and lays it flat.

7.2 Make the first fold

- The **right gripper** pinches the front corner of the throw's **head-side long edge** and the **left gripper** pinches the back corner of the same edge.
- Lift that edge just off the bed and carry it level toward the foot, one third of the way across the throw.
- Lay it down flat and let go. The **right gripper** smooths the new fold once with the closed face of the gripper.

Check: the folded edge runs straight from the front of the bed to the back, parallel to the long edges. The throw is now two layers thick over that third. If the edge lands crooked, both grippers lift it again and lay it down straight.

7.3 Make the second fold

- The **right gripper** pinches the front corner of the throw's **foot-side long edge** and the **left gripper** pinches the back corner of the same edge.
- Lift that edge just off the bed and carry it level toward the head, over the first fold, so it lands on top of it.
- Lay it down flat and let go. The **right gripper** smooths the new fold once with the closed face of the gripper.

Check: the throw is in thirds. It is three layers thick, one third as wide as it started, both folds run the same way, and both edges lie flat. If an edge sticks out past the fold, the gripper on that side lifts it and lays it down even.

Expected state: the throw lies folded in thirds on the foot half, running from the front edge of the bed to the back edge. Both pillows still stand at the headboard.

Step 8: Drape the throw at the foot

Goal: the folded throw lies across the foot of the bed, hanging the same over the front edge and the back edge.

8.1 Lift the folded throw

- The **right gripper** pinches the front end of the folded throw and the **left gripper** pinches the back end.
- Lift it just off the bed, keeping it level, so the folds do not open. Do not lift it high. Do not let an end swing down and touch the floor.

8.2 Lay it across the foot

- Carry it in a low carry toward the foot until it is over the foot end of the bed.
- Lay it down flat across the bed, its long edge running parallel to the foot edge, with the same amount hanging over the front edge and the back edge.
- Both grippers let go at the same time, so it lands flat and does not slide.

Check: the throw lies flat across the foot. Its long edge is straight. It has not opened out of its thirds. The bedding under it has not been dragged. If it lands rumpled, both grippers lift it again and lay it down flat.

Expected state: the folded throw lies across the foot of the bed. Both pillows still stand at the headboard.

Step 9: Square the corners

Goal: the throw is square on the bed and hangs the same over the front edge and the back edge.

9.1 Even the two sides

- Look at the front hang and the back hang of the throw from the front.
- The **right gripper** pinches the front end and the **left gripper** pinches the back end. Slide the throw a little toward the side that hangs short, until the same amount hangs on both sides.
- Both let go and come back and clear.

9.2 Straighten the edges

- The **right gripper** runs its closed face once along the throw's long edge nearest the foot, so the edge lies straight and parallel to the foot edge of the bed.
- The **right gripper** presses the front hanging corner flat, so it hangs straight down and does not curl, then reaches over and presses the back hanging corner flat the same way.

Check: the throw is square. Its long edge runs parallel to the foot edge, the same amount hangs over the front and the back, and neither hanging corner is curled or turned up. The throw is still in thirds. If one side hangs longer, both grippers slide it back the other way and check again.

Expected state: the throw lies square at the foot. Both pillows stand at the headboard. The bed top between them is clear and flat.

Step 10: End the episode

Goal: the head and the foot of the bed are dressed, and both arms are safely home.

- Confirm the first pillow is cased, full and even, and standing upright at the back half of the headboard.
- Confirm the second pillow is cased, full and even, and standing upright beside the first at the front half.
- Confirm the pair: both the same height, short edges touching, the same gap to the back edge and the front edge of the bed.
- Confirm the throw is folded in thirds, lying flat across the foot, square, hanging the same on both sides.
- Confirm the sheet and the blanket are still flat and tucked, and nothing has fallen on the floor.
- Confirm the base has not moved: it is still square to the bed and still locked.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded. The base stays parked and locked through the reset, and is only pushed away by hand once the bed is set for the next episode.

1. Lift both pillows off the headboard and lay them flat on the bed.
2. Take each pillow out of its case. Lay the two bare pillows flat, one on top of the other, at the pillow spot the next episode's config puts them: head half for Config L, front of the middle for Config M, foot half for Config R.
3. Turn each case right side out. Fold it flat and lay it at the case spot, one on top of the other, with the mouths pointing at the pillow spot: toward the head for Config L, toward the foot for Config M and R.
4. Lift the throw off the foot, shake it out, fold it in half once, and lay it at the throw spot, the back corner of the foot end, folded edge toward the head.
5. Smooth the sheet and the blanket. Tuck them again if the work pulled them loose.
6. Replace a pillow that stays lumpy after plumping, that will not spring back, or that is torn, damp, or stained.
7. Replace a case that is torn, damp, stained, or too small for the pillow.
8. Replace the throw if it holds a hard crease that will not lie flat, or if it is torn, damp, or stained.
9. Wash any item that touched the floor. Do not use it again this session.
10. Check the headboard is still fixed and does not rock, and clear the floor around both ends of the bed.
11. Check the base is still square, centered, and locked, and push it firmly once by hand to confirm it does not roll. If it has moved, park it again by the Base positioning steps.
12. Run the Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the pillow spot: the violations below were written for Config L (pillows in the head half, fed by the left gripper while the right gripper holds the mouth). The arm-role cues in them will be rewritten later to cover all three configs; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the grippers do does not match the config on the bed. The gripper on the far side of the bed reaches across to feed a pillow; the mouth is held by the gripper on the pillows' side; the pillows do not lie where that config puts them; the case mouths point the wrong way; or the wrong IF line is followed in Step 1 or 4.
- **SOP rule broken:** Steps 1.1 and 4.1, the same-side rule. The gripper on the pillows' side feeds, the other holds the mouth, and the IF line followed is the one for the config on the bed.
- **Coaching note:** look where the pillows lie before the first case is touched, then follow that config's IF lines through Steps 1 and 4.

Violation: Base moved during the episode

- **Visible cue:** the bed shifts in the frame, the side edge changes angle in the frame, or the base rolls, creeps, or turns at any time after recording starts.
- **SOP rule broken:** Step 10 and the Base positioning rule it confirms, the base is parked and locked before recording and does not move at all during the episode.
- **Coaching note:** park it, lock it, push-test it. If the park is wrong, fix it before recording, never during.

Violation: Done in the wrong order

- **Visible cue:** the second case is opened before the first pillow stands at the headboard, the throw is opened before both pillows stand at the headboard, or a pillow is touched again after the throw is opened.
- **SOP rule broken:** Steps 1 to 9, first pillow, second pillow, then the throw, one thing at a time.
- **Coaching note:** head of the bed first, foot of the bed last. Once the throw is open, the pillows are finished.

Violation: Pillow not fully cased

- **Visible cue:** bare pillow shows at the case mouth, the pillow sticks out of the mouth, or the case is left turned inside out.
- **SOP rule broken:** Steps 1.2, 1.3, 4.2, and 4.3, push the pillow all the way down the case and pull the mouth back over it.

- **Coaching note:** all the way in, then pull the case down. If the pillow can be seen, it is not cased.

Violation: Case corners left empty

- **Visible cue:** one or both far corners of the case are flat and empty, so the case has an ear hanging past the pillow.
- **SOP rule broken:** Steps 1.3 and 4.3, the mouth gripper presses each far corner of the pillow down into its own corner of the case.
- **Coaching note:** push the corners home before the case is smoothed. An empty corner shows from across the room.

Violation: Pillow not plumped

- **Visible cue:** the pillow goes to the headboard flat, with a hollow in the middle or a hard lump at one end, and no press is seen in the episode.
- **SOP rule broken:** Steps 2 and 5, press the ends twice and even out the lumps before the pillow leaves the case spot.
- **Coaching note:** press, let go, press again. A flat pillow at the headboard is the first thing anyone sees.

Violation: Plumped in the air

- **Visible cue:** a gripper lifts the pillow off the bed and presses, shakes, or punches it while it hangs.
- **SOP rule broken:** Steps 2.1 and 5.1, plump the pillow while it lies flat on the case spot.
- **Coaching note:** the bed holds it while it is pressed. Nothing is worked hanging in the air.

Violation: Pillow not standing right

- **Visible cue:** a pillow leans away from the headboard, falls forward when the left gripper lets go, lies flat instead of standing, or sits low.
- **SOP rule broken:** Steps 3.2 and 6.2, stand the pillow upright against the headboard with its bottom edge on the bed and its face leaning on the headboard.
- **Coaching note:** stand it, then watch it settle before letting go. A pillow that falls is a pillow that was not finished.

Violation: Pillows not matched

- **Visible cue:** at the end the two pillows stand at different heights, their short edges do not touch, the first pillow is at the front half instead of the back, or the gap to the back edge and the front edge of the bed is clearly different.
- **SOP rule broken:** Steps 3.2 and 6.2, the first pillow goes to the back half, the second beside it at the front half, same height, edges touching, gaps even.

- **Coaching note:** look at the pair from the front, not at one pillow at a time. Back half first, so the first is never in the way of the second.

Violation: Throw not folded in thirds

- **Visible cue:** the throw ends up folded in half, folded in quarters, folded the wrong way, or with an edge sticking out past a fold.
- **SOP rule broken:** Steps 7.2 and 7.3, two folds the same way, three layers, one third as wide.
- **Coaching note:** one third over, then the other third on top. Count the layers before lifting it.

Violation: Throw opened out during the carry

- **Visible cue:** a fold falls open while the throw is carried to the foot, or the throw is dragged along the bed instead of lifted level by both ends.
- **SOP rule broken:** Step 8.1, lift it level by both ends so the folds stay shut.
- **Coaching note:** level by both ends. A tilted throw unfolds itself.

Violation: Throw not square

- **Visible cue:** at the end the throw hangs clearly longer over one edge than the other, its long edge runs at an angle to the foot edge, or a hanging corner is curled or turned up.
- **SOP rule broken:** Step 9, even the two sides, straighten the long edge, and press the hanging corners flat.
- **Coaching note:** check both sides from the front. Square means the same at the front and the back.

Violation: Bedding pulled or bunched

- **Visible cue:** the sheet or the blanket is dragged, rumped, bunched, or pulled out of its tuck by an arm, a carry, or a press.
- **SOP rule broken:** Steps 1 to 9, the bedding under the work never moves.
- **Coaching note:** lift things, do not drag them. The bed that was given is the bed that is left.

Violation: Dragged instead of lifted

- **Visible cue:** a pillow, a case, or the throw is slid or pushed along the bed top instead of lifted straight up before it moves.
- **SOP rule broken:** Steps 1 to 9, every pick lifts straight up, just off the bed, before it moves anywhere.
- **Coaching note:** up first, then across. Dragging rumples everything under it.

Violation: Lifted high or over the headboard

- **Visible cue:** a pillow, a case, or the throw is lifted well above the bed instead of just off it, or is carried over the top of the headboard.

- **SOP rule broken:** Steps 3.1, 6.1, and 8.1, lift just off, then carry low and level over the bed top.
- **Coaching note:** up a little, then along. There is no reason to go high.

Violation: Two things moved at once

- **Visible cue:** both grippers carry different things at the same time, or a thing is passed from one gripper to the other.
- **SOP rule broken:** Steps 1 to 9, one thing moves at a time, and nothing is passed between grippers.
- **Coaching note:** finish one thing and put it down before the next one starts.

Violation: Leaned on or moved the headboard

- **Visible cue:** a gripper, wrist, or forearm rests on the headboard, or the headboard rocks, slides, or is pushed by a pillow set down too hard.
- **SOP rule broken:** Steps 3.2 and 6.2, the headboard is fixed, nothing leans on it or pushes it.
- **Coaching note:** the arm holds itself up. Lean the pillow on the headboard, not the arm.

Violation: Wrong arm used

- **Visible cue:** the right gripper carries a pillow to the headboard or stands it up, the left gripper takes the front corner or the front end of the throw, or the right gripper takes the back corner or the back end.
- **SOP rule broken:** Steps 3, 6, 7, 8, and 9, the left gripper owns the headboard, the right gripper takes the front of the throw and the left gripper the back.
- **Coaching note:** left arm to the headboard. Right arm at the front of the throw, left arm at the back.

Violation: Something dropped

- **Visible cue:** a pillow, a case, or the throw slips out of a gripper, falls off the bed, or touches the floor.
- **SOP rule broken:** Steps 1 to 9, nothing goes on the floor, and each thing is put down under control.
- **Coaching note:** close all the way before lifting, and put it down before letting go. Report a floor touch, do not carry on.

Violation: Failed check not retried

- **Visible cue:** a check in a step clearly fails and the episode carries on with no retry: a case corner is left empty, a pillow falls forward, a fold lands crooked, or the throw hangs long on one side.

- **SOP rule broken:** Steps 1 to 9, do the retry written under each failed check before moving on.
- **Coaching note:** a check is only worth doing if the retry is done. Fix it where it happened.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a pillow uncased, flat, or lying down, the throw unfolded, off the foot, or hanging uneven, the bedding pulled, or an arm away from home.
- **SOP rule broken:** Step 10 (confirm both pillows, the pair, the throw, the bedding, the floor, and the locked base; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot see the headboard, the foot edge, or both side edges**, so nobody can tell if the pillows stand right or if the throw is square.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake that releases on its own, or a caster that seizes so the base cannot be parked square.
- **Defective object:** a pillow that stays lumpy after a correct plump, will not spring back, or arrives torn, damp, or stained; a case that arrives torn, damp, stained, sewn shut, or too small to take its pillow; a throw that arrives torn, damp, stained, too short to hang on both sides, or creased so hard it will not lie flat. Replace it before the next episode.
- **Bed fault:** a bed that rocks, a headboard that is loose or falls forward, or bedding that arrives untucked or wet before the episode starts.
- **Reach fault:** a place turns out to be too far for its arm with the base parked correctly, so the pillows, the cases, the throw, the headboard end, or the foot end cannot be reached without extending the arm to a limit.

Annotation subtasks (from SOP)

1. Open the first case mouth
2. Feed the first pillow into its case
3. Fill the case corners and settle the case
4. Plump the first pillow
5. Carry the first pillow to the headboard and stand it at the back half
6. Open the second case mouth

7. Feed the second pillow into its case
8. Fill the case corners and settle the case
9. Plump the second pillow
10. Carry the second pillow to the headboard and stand it beside the first
11. Open the throw out flat
12. Make the first fold
13. Make the second fold
14. Lift the folded throw and lay it across the foot of the bed
15. Even the two hanging sides
16. Straighten the edge and press the corners
17. Return both arms home and end the episode